Technology Stack

Date	01 July 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Prediction Technology Stack:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Backend / Server-Side	Python (Flask)	Flask is a lightweight Python framework used to handle user requests, process input data, load the trained machine learning model, and generate predictions.
2	Frontend / Client-Side	HTML, CSS, JavaScript	Used to build a responsive and user-friendly web interface where users can enter development indicators and view the predicted HDI score and category

3	Database / Data Storage	CSV Dataset (HDI.csv)	Stores Human Development Index data used for model training and analysis. The CSV format is simple, portable, and suitable for academic projects.
4	Cloud / Hosting / Deployment	Render	Render provides easy deployment, automatic builds, scalability, and cost-effective hosting for web applications.
5	Version Control & CI/CD	GitHub	GitHub enables version control, code management, collaboration among team members, and project tracking.
6	Third-Party APIs / Other Tools	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Pickle, Jupyter Notebook	Pandas and NumPy are used for data processing, Scikit-learn for machine learning model development, Matplotlib and Seaborn for visualization, Pickle for model serialization, and Jupyter Notebook for experimentation and model training.